SHOC SOPs 6

Information Communication and Technology (ICT) SOPs

Data governance, infrastructure resilience, cybersecurity, and digital reporting mechanisms.

SADC Humanitarian Emergency Operations Centre, Nacala, Mozambique

SADC HUMANITARIAN AND EMERGENCY OPERATIONS CENTRE

STANDARD OPERATING PROCEDURE

ICT, Data Management and Reporting

SOP Data Management | SOP ICT Infrastructure | SOP Assessment & Reporting
SOP BCM & Disaster Recovery | SOP Cybersecurity Ops | SOP Steady-State ICT

Structured per Mission Cycle

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1. Purpose, Scope and Legal Basis

1.1 Purpose

This Volume establishes the integrated ICT, data management, reporting, business continuity, cybersecurity, and steady-state operations framework for SHOC. It consolidates six interdependent SOPs into a single operational document structured per the UNDAC Mission Cycle: SOP 6.1 (Data Management) governs data collection, validation, classification, storage, and access control; SOP 6.2 (ICT Infrastructure) provides the technology platform enabling all SHOC operations; SOP 6.3 (Assessment, Data Analysis and Reporting) produces decision-support products; SOP 6.4 (BCM and Disaster Recovery) ensures organisational ICT continuity and failover; SOP 6.5 (Cybersecurity Operations) provides continuous security posture management; SOP 6.6 (Steady-State ICT Operations) governs day-to-day ICT administration. Together, they ensure SHOC's ICT resilience extends beyond crisis response to permanent organisational capability.

1.2 Scope and Applicability

Applies to SHOC ICT and IM Units, all staff (positions 1–23), secondees, surge personnel, SADC Member State NDMAs exchanging data with SHOC, and all partners accessing SHOC systems. Covers SHOC HQ (Nacala), Botswana OSOCC, Virtual OSOCC environments, and field deployment sites across all 16 SADC Member States.

Cross-Functional Data Coverage: Finance (SOP 1), Early Warning (SOP 2), Emergency Communications (SOP 3), Emergency Response (SOP 4), HR (SOP 5), Security (SOP 7), Supply Chain (SOP 8), BCM (SOP 9).

1.3 Legal and Institutional Basis

Primary: SHOC MOA Article 6(d–e); SADC DRR Strategy (2022–2030); SADC Digital Transformation Strategy; Sendai Framework Priority 1.

Standards: ISO 27001:2022 (Information Security); ISO 22301:2019 (BCM); ITU Radio Regulations; WFP ETC/TESS+ Standards; IFRC Surge Toolbox IT Module.

SADC Regional Instruments: SADC Model National Emergency Telecommunications Plan (NETP, October 2023); SADC Framework for Harmonisation of Radio Frequency for PPDR (2021); Tampere Convention on the Provision of Telecommunication Resources for Disaster Mitigation and Relief Operations (1998); SADC/CRASA Emergency Services Numbering harmonisation; EW4All implementation framework.

Reference: UNDAC Handbook 8th Ed. (2024); SHOC ICT Policy v3.0; SHOC BCM Plan (SOP 9.1–9.7); SHOC Telecom SOP v3.0 (emergency telecommunications and field deployment).

1.4 Guiding Principles

a) Data integrity, security, and ethical use per ISO 27001 and UNDAC data ethics

b) Decision-grade data through systematic validation and quality assurance

c) Zero-loss continuity with defined RPO/RTO targets per BCM Plan

d) Interoperability with NDMA, SADC, AU, and UN humanitarian platforms

e) Five-tier data classification (Public/Restricted/Confidential/Secret/Top Secret)

f) Role-based access control aligned to SHOC staffing model (positions 1–23)

g) Technology resilience through hybrid satellite–cloud continuity architecture

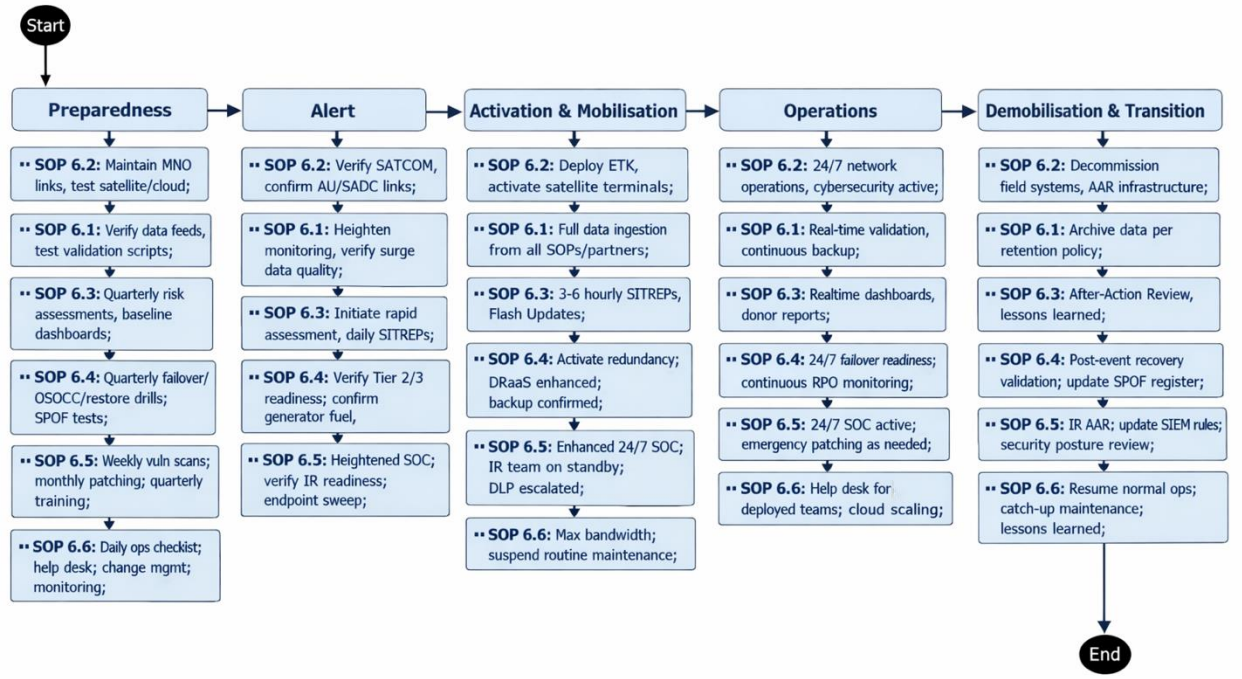
h) PPDR alignment: SADC harmonised frequency bands and Model III commercial network architecture with satellite backup, per SADC PPDR Framework (2021)

i) Tampere Convention compliance: pre-prepared regulatory barrier reduction for cross-border equipment deployment across all 16 SADC Member States

j) Accessibility and inclusion: public warning and reporting outputs accessible to persons with disabilities, the elderly, and vulnerable groups per NETP Action Item 10 and EW4All Pillar 3

2. Operational Cycle for ICT and Data

Each phase in the Emergency defines specific ICT, data management, and reporting actions.



Phase Summary

Phase	SOP 6.1 Data Mgmt	SOP 6.2 ICT Infra	SOP 6.3 Reporting
Preparedness	Verify feeds; test validation; metadata baselines; annual training	Monitor MNO performance; monthly SATCOM tests; maintain VPNs; patch mgmt	Quarterly risk assessments; baseline dashboard updates; monthly summaries
Alert	Increase monitoring; verify surge data quality; heightened QA	Verify Vodacom/Movitel capacity; pre-check satellite; alert AU/SADC	Rapid assessment initiation; daily SITREPs; 6-hour dashboards
Activation & Mobilisation	Full data flow from all SOPs; real-time ingestion; partner feeds active	Deploy ETK; activate Starlink/VSAT; full SATCOM; ETC coordination	3–6 hourly SITREPs; Flash Updates; real-time dashboards; gap analysis
Operations	Real-time validation and QA; continuous cloud backup; 24/7 data flow	24/7 network ops; enhanced cybersecurity; maximum bandwidth	Real-time dashboards; daily financial SITREPs; donor reports; coordination
Demobilisation & Transition	Archive per retention policy; secure disposal; lessons learned	Decommission field systems; equipment return; infrastructure AAR	After-Action Review within 30 days; update risk register; archive reports

Phase Summary: SOPs 6.4–6.6 (BCM, Cybersecurity, Steady-State)

Phase	SOP 6.4 BCM & DR	SOP 6.5 Cybersecurity	SOP 6.6 Steady-State
Preparedness	Quarterly failover/OSOCC/restore drills; monthly power tests; SPOF verification	Weekly vulnerability scans; monthly patching; quarterly training; SIEM tuning	Daily ops checklist; help desk; change mgmt; network/cloud monitoring; asset lifecycle
Alert	Verify Tier 2/3 readiness; confirm generator fuel; pre-stage failover	Heightened SOC; verify IR readiness; endpoint compliance sweep	Verify all systems green; clear help desk backlog; freeze non-emergency changes
Activation & Mobilisation	Activate redundancy; DRaaS enhanced; confirm backup currency	Enhanced 24/7 SOC; IR team on standby; DLP monitoring escalated	Maximum bandwidth; suspend routine maintenance; all resources to operations
Operations	24/7 failover readiness; continuous backup verification; RPO monitoring	24/7 SOC active; real-time threat monitoring; emergency patching as needed	Help desk for deployed teams; network performance monitoring; cloud scaling
Demobilisation & Transition	Post-event recovery validation; AAR; update SPOF register; BCM exercise	IR AAR; update SIEM rules; security posture review; training updates	Resume normal ops; catch-up maintenance; change freeze lifted; lessons learned

3. Governance, Roles and RACI Matrix

3.1 Consolidated RACI Matrix

R=Responsible, A=Accountable, C=Consulted, I=Informed

Function	R	A	C	I
Data Collection & Validation (6.1)	IM Officer	ICT Officer	EW Officer, All Units	NDMAs, Partners
Data Classification (6.1)	ICT Officer	Head of Ops	Legal Advisor	All Staff
Database & Backup (6.1)	Asst ICT	ICT Officer	Cloud Provider	Head of Ops
Access Control (6.1/6.2)	ICT Officer	ED	Security Officer	All Users
Cloud Infrastructure (6.2)	ICT Officer	Head of Ops	Cloud Provider	All Staff
Local Network / MNO (6.2)	Asst ICT	ICT Officer	Vodacom/Movitel	SHOC Teams
Satellite Communications (6.2)	ICT Officer	Head of Ops	SATCOM Provider	NDMAs
Regional/AU Connectivity (6.2)	ICT Officer	ED	SADC IT, AU PAPS	Partners
Cybersecurity (6.2)	ICT Officer	ED	SOC Provider	All Users
ETK Deployment (6.2)	ICT Officer	Ops Officer	ETC, Vendors	ERTs
Assessment Coordination (6.3)	IM Officer	Head of Ops	EW Officer	NDMAs
SITREP Production (6.3)	IM Officer	Head of Ops	All Units	ED, NDMAs

Dashboard Management (6.3)	ICT Officer	IM Officer	All Units	Leadership
Donor Reporting (6.3)	Finance Officer	ED	IM Officer	Donors
BCM Failover Activation (6.4)	ICT Officer	ED	Head of Ops	All Units
DR Testing & Restore Drills (6.4)	ICT Officer	Head of Ops	Asst ICT	All Units
Incident Response (6.5)	ICT Officer	Head of Ops	SECaaS Provider	ED, All Units
Vulnerability Mgmt & Patching (6.5)	ICT Officer	Head of Ops	Asst ICT	All Units
Security Training (6.5)	ICT Officer	Head of Ops	HR Officer	All Staff
Daily ICT Operations (6.6)	Asst ICT	ICT Officer	—	Head of Ops
Help Desk & User Support (6.6)	Asst ICT	ICT Officer	—	All Staff
Change Management (6.6)	ICT Officer	Head of Ops	Affected Unit Head	All Units
Asset Lifecycle & Licensing (6.6)	ICT Officer	Head of Ops	Finance Officer	ED

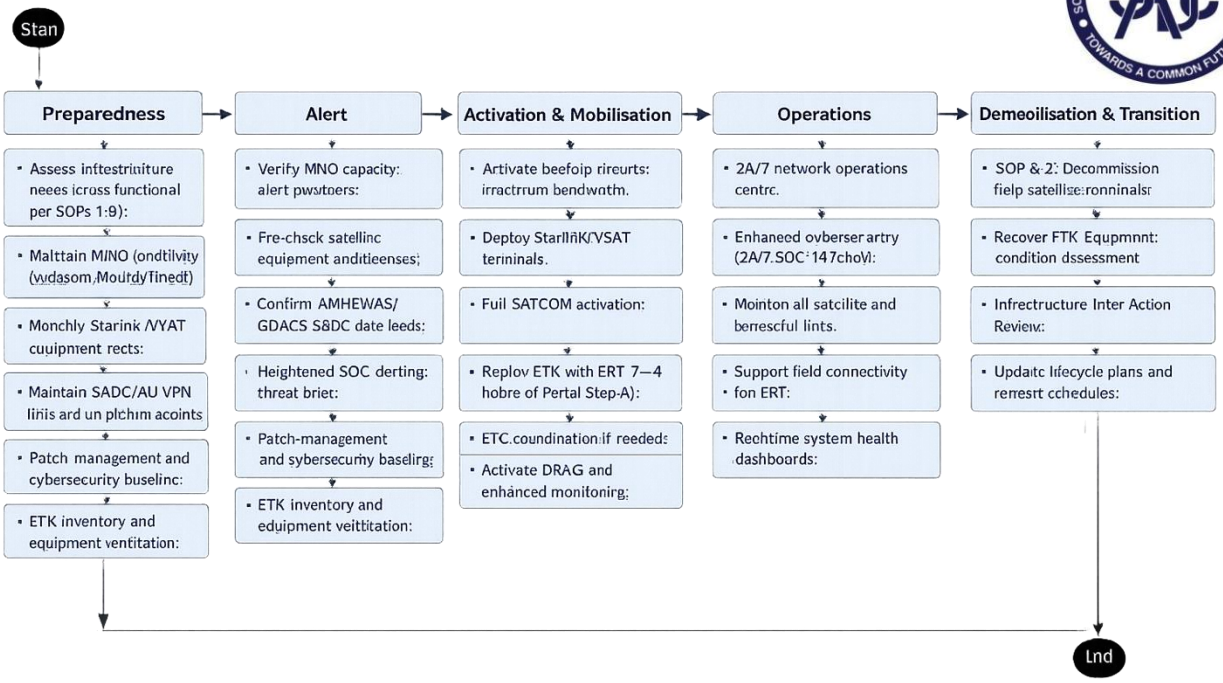
3.2 Data Stewardship

Each functional area has a designated Data Steward responsible for data accuracy within their domain: Finance → Finance & Admin Officer; Early Warning → EW Officer; Communications → Comms Officer; Emergency Response → Operations Officer; HR → HR Officer; ICT → ICT Officer; Security → Security Officer; Procurement → SCM Officer; BCM → ICT Officer.

4. SOP 6.2: ICT Infrastructure and New Technologies

Purpose: Deploy, manage and optimise SHOC's technology ecosystem — cloud, satellite, terrestrial networks, cybersecurity, and ETK — providing the platform upon which SOPs 6.1 and 6.3 depend. Located in Nacala, Mozambique, SHOC operates within a specific telecommunications context requiring integration with local MNOs (Vodacom, Movitel, Tmcel), SADC infrastructure, AU continental systems, and global UN platforms.

Linked Annexes: 6.2-A (MNO Contact Directory), 6.2-B (Satellite Equipment Inventory), 6.2-C (VPN Config Standards), 6.2-D (Cloud Evaluation Matrix), 6.2-E (Security Baseline – TESS+), 6.2-F (AU/UN Credentials Register), 6.2-G (MET Feed Specs), 6.2-H (ETC Procedures), 6.2-I (ETK Checklist), 6.2-J (Incident Response), 6.2-K (Network Topology), 6.2-L (Cross-Functional ICT Matrix)



4.1 Preparedness

#	Action	Responsible	Output	Threshold	Annex Ref
1	Assess cross-functional ICT needs per SOPs 1–9; update requirements matrix	ICT Officer	ICT Requirements Matrix	Annual/Q4	6.2-L
2	Monitor MNO performance (Vodacom/Movitel/Tmcel); test backup SIMs	Asst ICT	MNO Status Report	Monthly	6.2-A
3	Monthly Starlink/VSAT/BGAN equipment tests; verify satellite licenses	ICT Officer	SATCOM Test Report	Monthly	6.2-B
4	Maintain SADC/AU VPN links; test AMHEWAS/GDACS/VOSOCC connectivity	ICT Officer	VPN Status Log	Monthly	6.2-C, 6.2-F
5	Cybersecurity baseline: patch management, SIEM monitoring, penetration tests	ICT Officer	Security Report	Quarterly	6.2-E
6	ETK inventory verification; equipment functionality checks	Asst ICT	ETK Inventory Report	Monthly	6.2-I

4.2 Alert

Verify Vodacom/Movitel capacity and alert providers. Pre-check satellite equipment and confirm licenses. Confirm AMHEWAS/GDACS data feeds. Heightened SOC alerting with threat brief. Pre-position ETK if required. Verify financial systems (ERP) access and alert Finance unit. Confirm HR systems and surge roster readiness.

4.2A SADC PPDR Frequency Bands and Network Architecture

Per the SADC Framework for Harmonisation of Radio Frequency for PPDR (2021), SHOC emergency telecommunications equipment shall be aligned to the following harmonised SADC PPDR frequency bands. The Frequency Pre-Clearance Cache (maintained under SHOC Telecom SOP v3.0) includes pre-arranged clearances for these bands across all 16 SADC Member States.

PPDR Type	Frequency Bands	Application
Narrowband	380–385 MHz paired with 390–395 MHz; 410–420 MHz paired with 420–430 MHz	Voice communications; low-data telemetry; first responder radio (TETRA, DMR, P25)
Wideband	380–470 MHz (preferably 380–430 MHz)	Data services; video; enhanced voice; sensor telemetry
Broadband	698–703 MHz (UL) / 753–758 MHz (DL); 733–736 MHz (UL) / 788–791 MHz (DL)	Broadband data; real-time video; IoT; high-throughput field connectivity

PPDR Network Model: SHOC operates under PPDR Model III (Commercial Network) as defined in the SADC PPDR Framework, obtaining emergency services from commercial MNO providers (Vodacom, Movitel, Tmcel) supplemented by satellite backup (Starlink, VSAT, BGAN). This deliberate architecture choice reflects the current SADC telecommunications infrastructure maturity and enables rapid deployment without requiring dedicated PPDR network investment. As regional broadband PPDR networks mature, SHOC may transition to Model II (Shared Network) with priority access and pre-emption rights for emergency use. This transition pathway shall be reviewed annually with the SADC Secretariat ICT Division.

PPDR Equipment Standards: ETK and field-deployed radio equipment shall preferentially conform to TETRA (Terrestrial Trunked Radio), DMR (Digital Mobile Radio), or P25 standards to maximise cross-border interoperability across SADC Member States. Where SHOC procures new HF/VHF/UHF equipment, specifications shall reference SADC PPDR harmonised bands and interoperable standards. Equipment procurement shall follow ETC/IFRC ERU specifications with SADC PPDR compatibility as a weighted evaluation criterion. The SHOC Telecom SOP v3.0 Frequency Pre-Clearance Cache includes regulatory contact databases and customs facilitation packs for all 16 SADC Member State spectrum authorities, Tampere Convention documentation packages, and template frequency allocation requests.

4.3 Activation and Mobilisation

#	Action	Responsible	Output	Threshold	Annex Ref
1	Activate backup circuits; scale to maximum bandwidth	ICT Officer	Bandwidth Confirmation	≤2 hours	6.2-K
2	Deploy Starlink/VSAT terminals for field operations	ICT Officer	SATCOM Active	≤4 hours	6.2-B
3	Deploy ETK with ERT (per SOP 3.1/4.1 Portal Step 4)	ICT/Ops Officer	ETK Deployed	≤4 hours	6.2-I
4	Activate DRaaS; enhanced cloud monitoring; confirm failover readiness	Asst ICT	DRaaS Active	≤2 hours	6.2-D

5	Coordinate with WFP ETC if needed; activate Crisis Connectivity Charter	ICT Officer	ETC Coordination Log	As needed	6.2-H

4.4 Operations

24/7 network operations. Enhanced cybersecurity with SOC active and incident response ready. Monitor all satellite and terrestrial links continuously. Support field connectivity for deployed ERTs. Real-time system health dashboards operational. RTO targets: EW systems 0 hrs; Operations 1 hr; IM/SITREP 1 hr; Communications 4 hrs; Finance 12 hrs.

4.5 Demobilisation and Transition

Decommission field satellite terminals; recover and inspect ETK equipment. Conduct infrastructure After-Action Review. Update technology lifecycle plans, refresh schedules, and vendor SLAs. Archive system logs per SOP 6.1 retention policy.

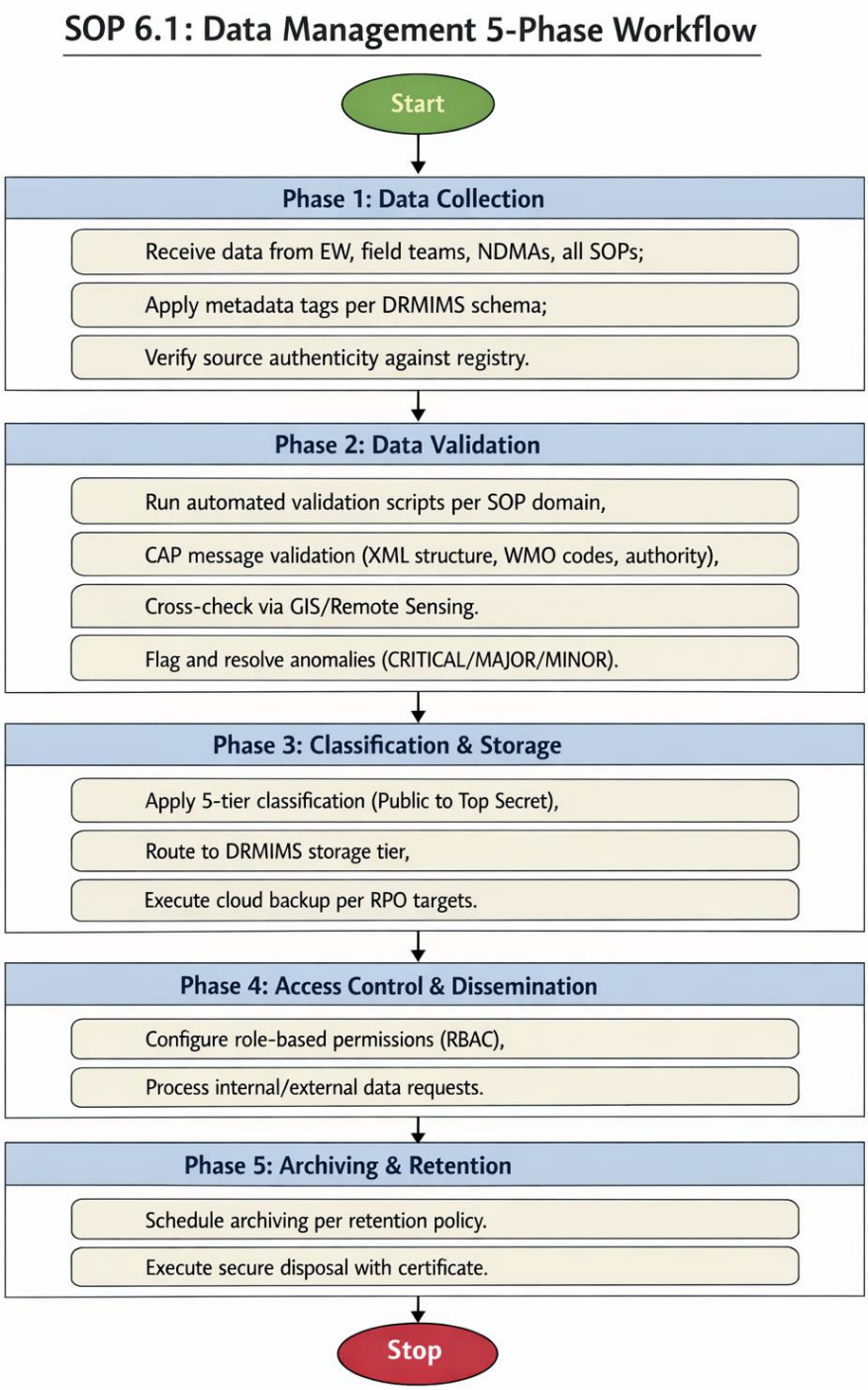
5. SOP 6.1: Data Management

Purpose: Establish standardised data collection, validation, classification, storage, access control, and archiving procedures for all SHOC information assets. Operates on the infrastructure provided by SOP 6.2 and feeds decision-grade data to SOP 6.3 for analysis and reporting. Data flows through DRMIMS as the primary repository, with hybrid satellite–cloud backup architecture.

Linked Annexes: 6.1-A (Data Receipt Register), 6.1-B (Metadata Schema), 6.1-C (Classification Matrix), 6.1-D (Data Request Register), 6.1-E (Backup Verification Checklist), 6.1-F (Data Quality Form), 6.1-G (Exchange Specifications), 6.1-H (Cross-SOP Data Flow), 6.1-I (Stewardship Matrix), 6.1-J (AAR Template)

5.1 Data Classification Scheme

Classification	Description	Handling	Encryption	Retention
Public	Press releases, general advisories	No restrictions	Optional	Permanent
Restricted	Internal operational info, daily reports	SHOC staff only	In transit	5 years
Confidential	HR records, finance data, assessments	Need-to-know; encrypted	At rest + transit	7 years
Secret	Security/strategic information, threats	Restricted access; audit trail	At rest + transit	10 years
Top Secret	Critical intelligence, classified ops	ED approval; enhanced controls	Enhanced	Permanent



5.2 Preparedness

#	Action	Responsible	Output	Threshold	Annex Ref
1.1	Verify all data ingestion feeds (DRMIMS, APIs, satellite, AMHEWAS, GDACS)	IM Officer	Feed Status Report	Monthly	6.1-G
1.2	Test validation scripts per SOP domain; update business rules	IM Officer	Script Test Report	Quarterly	6.1-F

1.3	Review data classification policies; conduct annual staff training	ICT Officer	Training Record	Annual	6.1-C
1.4	Verify backup systems and RPO compliance; test recovery procedures	Asst ICT	Backup Test Report	Monthly	6.1-E

5.3 Activation and Mobilisation / Operations

#	Action	Responsible	Output	Threshold	Annex Ref
2.1	Full data ingestion from all SOPs and external partners; real-time feeds active	IM Officer	Raw Data Ingested	≤1 hour	6.1-A
2.2	Apply metadata tags; standardise formats per DRMIMS schema	IM Officer	Metadata Record	≤2 hours	6.1-B
2.3	Run automated validation; GIS cross-check for spatial data	IM Officer	Validation Report	≤6 hours	6.1-F
2.3A	CAP message validation: verify CAP XML structure compliance, hazard type codes against WMO/CAP standards, geographic polygon within SADC bounds, urgency/severity/certainty fields, sender authority against Authorised Alerting Authority Register. Flag non-compliant CAP messages for ICT Officer review before dissemination. Cross-reference CAP alerts with AMHEWAS/GDACS feeds for consistency.	IM Officer	CAP Validation Log	≤30 min per msg	6.1-F, 6.3-E
2.4	Classify validated data per 5-tier scheme; route to storage tier	ICT Officer	Classification Tag	≤14 hours	6.1-C
2.5	Execute cloud backup per RPO (EW ≤15min; Ops 30min; Finance 1hr)	Asst ICT	Backup Confirmation	Per RPO	6.1-E
2.6	Process data requests; maintain RBAC; log all access	ICT Officer	Request Log	≤24 hours	6.1-D

5.4 Demobilisation and Transition

Archive data per retention schedule (5–10 years by classification). Apply legal holds where required. Execute secure disposal with certified destruction for expired data. Maintain permanent disposal log for audit purposes. Conduct data management lessons learned for AAR.

5.5 Storage Architecture

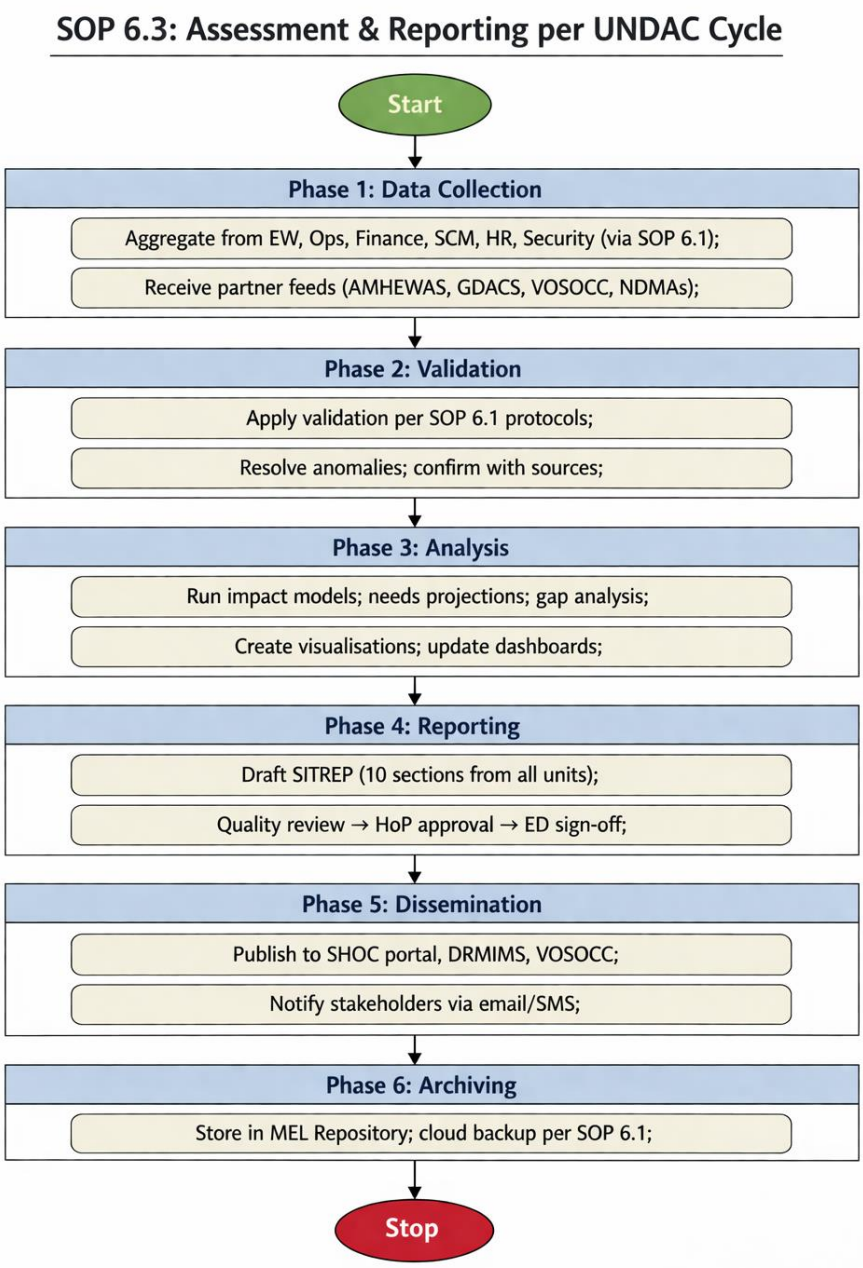
Primary (On-Site): SHOC HQ Nacala servers — active operational data. Cloud Primary (Azure SA Region A): real-time DRMIMS replication. Cloud Secondary (SADC Regional): geo-

redundant disaster recovery. Satellite Buffer (Botswana OSOCC): failover cache during connectivity loss. Cold Archive: immutable off-site for compliance archives (7+ years).

6. SOP 6.3: Assessment, Data Analysis and Reporting

Purpose: Conduct multi-hazard assessments, analyse operational data from all SHOC units, and generate decision-support reports (SITREPs, Flash Updates, Dashboards, Donor Reports). Consumes validated data from SOP 6.1, operates on SOP 6.2 infrastructure, and integrates inputs from all other SOP Volumes.

Linked Annexes: 6.3-A (SITREP Template), 6.3-B (Flash Update Template), 6.3-C (Rapid Needs Assessment Form), 6.3-D (Dashboard Indicator Definitions), 6.3-E (Data Quality Checklist), 6.3-F (Cross-SOP Data Flow), 6.3-G (AAR Template), 6.3-H (Donor Reporting Template), 6.3-I (Accessibility Checklist), 6.3-J (Call Centre Setup Checklist)



6.1 Reporting Framework

Report Type	Phase	Frequency	Audience	BCM Priority
SITREP	Activation–Operations	3-hr (crisis); 6-hr (activation)	All stakeholders	Critical
Flash Update	Alert–Operations	Event-triggered	Leadership, NDMAs	Critical
EW Bulletin	All phases	Daily (prep); hourly (crisis)	NDMAs, Partners	Critical
Financial SITREP	Activation–Operations	Daily (per SOP 1.2)	ED, Donors	High
Donor Report	Operations–Demob	Weekly; Final	Donors, Secretariat	Medium
AAR	Demobilisation	Within 30 days	All stakeholders	Medium
Dashboard Update	All phases	Real-time (crisis); daily (prep)	All users	Critical

6.2 Preparedness

#	Action	Responsible	Output	Threshold	Annex Ref
1	Conduct quarterly risk assessments; maintain baseline data	IM Officer	Risk Profile	Quarterly	6.3-C
2	Update dashboard indicators and visualisation templates	ICT/IM Officer	Dashboard Config	Quarterly	6.3-D
3	Verify SITREP template currency; train reporting staff	IM Officer	Template Package	Annual	6.3-A, 6.3-B

6.3 Alert / Activation and Mobilisation

#	Action	Responsible	Output	Threshold	Annex Ref
1.1	Aggregate data from all SOP domains and partner feeds	IM Officer	Raw Data Package	≤2 hours	6.3-F
2.1	Validate per SOP 6.1; GIS cross-check; resolve anomalies	IM Officer	Validated Dataset	≤6 hours	6.3-E
3.1	Run analytical models; impact projections; gap analysis	IM Officer	Analysis Report	≤8 hours	—
3.2	Create visualisations; update operational dashboards	IM/ICT Officer	Dashboard Update	≤2 hours	6.3-D
4.1	Draft SITREP (10 sections integrating all unit inputs)	IM Officer	Draft SITREP	≤3 hours	6.3-A
4.2	Quality review → HoP approval → ED sign-off	HoP / ED	Approved SITREP	≤1 hour	—

5.1	Publish to SHOC portal, DRMIMS, VOSOCC; notify stakeholders	IM Officer	Published Report	≤30 min	—

6.4 Demobilisation and Transition

Produce After-Action Review within 30 days integrating lessons from all units. Archive complete report packages in SHOC MEL Repository with verified cloud backup. Update risk register and assessment baselines. Recommend SOP revisions based on lessons learned.

6.5 Accessibility in Public Warning and Reporting (NETP AI 10)

Requirement: Per SADC Model NETP Action Item 10 and EW4All Pillar 3 (Warning Dissemination and Communication), all SHOC public-facing information products shall be accessible to persons with disabilities (PWDs), the elderly, women and girls, refugees, and immigrants. This applies to SITREPs, Flash Updates, EW Bulletins, dashboard outputs, and all public warning dissemination.

Provisions: (a) Text-based alerts shall use plain language at a maximum 8th-grade reading level, avoiding jargon and technical abbreviations. (b) Visual products (maps, dashboards, infographics) shall use high-contrast colour schemes accessible to colour-blind users and include text descriptions of key visual elements. (c) Audio alerts disseminated via radio/TV broadcasting shall include sign language interpretation or text captioning where broadcast infrastructure permits. (d) SMS/Cell Broadcast alerts shall be formulated in multiple SADC official languages (English, French, Portuguese) and, where feasible, local languages of the affected area. (e) SITREP distribution to NDMA partners shall include accessible PDF format (tagged PDF with reading order). (f) The IM Officer shall maintain an Accessibility Checklist (Annex 6.3-I) and verify compliance before each SITREP/Flash Update publication. (g) Accessibility compliance shall be reviewed annually with SADC Secretariat DRR Unit and reported in the KPI framework.

6.6 Emergency Call Centre Procedures (NETP AI 14)

Purpose: To establish temporary call centres during the Operations phase to warn affected populations of new risks, disseminate emergency updates, and connect affected populations with their relatives. Per SADC Model NETP Action Item 14, these call centres supplement digital reporting channels in areas where internet access is limited.

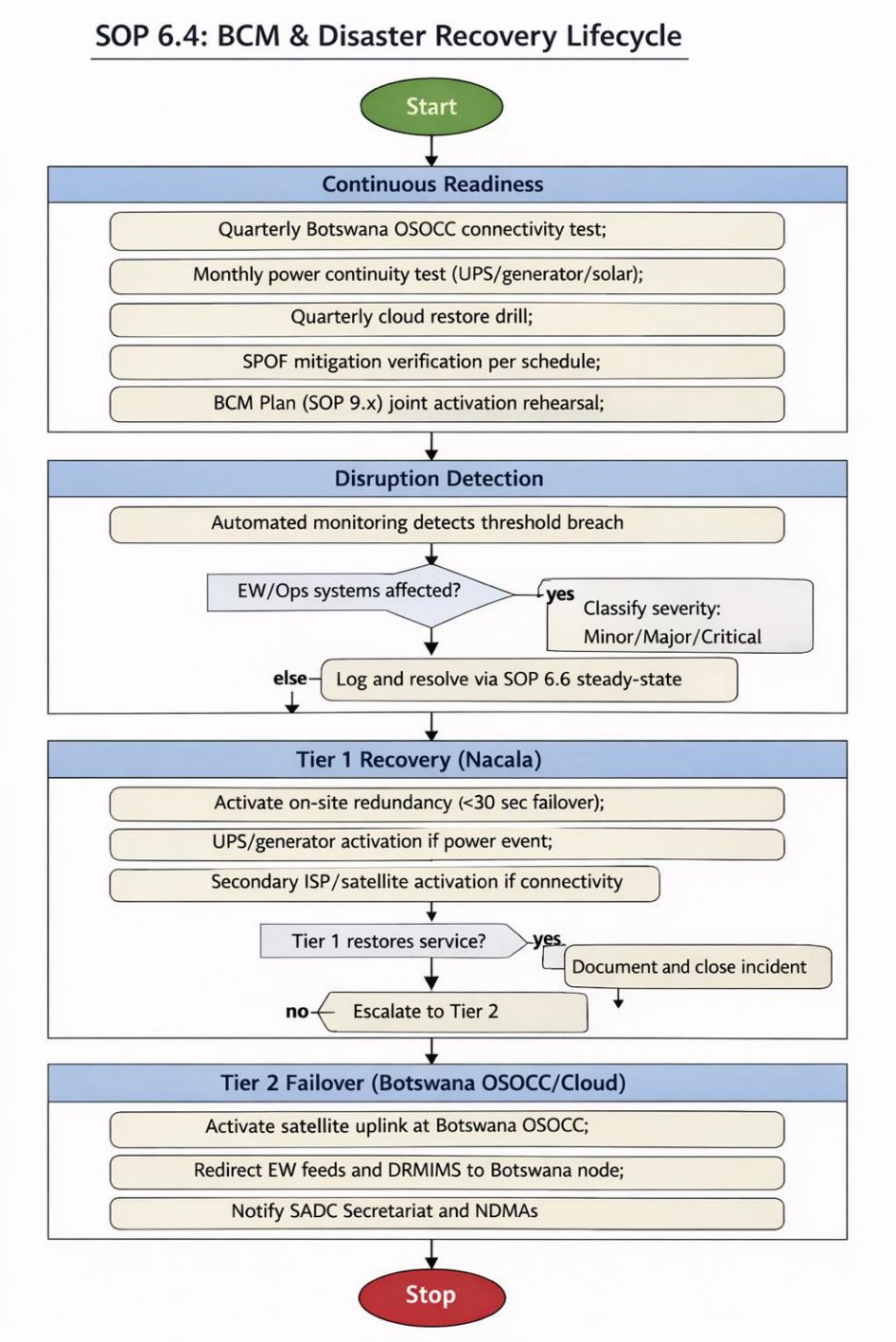
Activation: Call centre establishment is a mission-specific optional deployment, activated by the Head of Operations upon assessment of affected area communications capacity. Activation triggers: (a) terrestrial mobile network outage affecting >50% of disaster zone; (b) affected population >50,000 with limited digital access; (c) NDMA request for population communication support.

Procedures: (1) ICT Officer coordinates satellite connectivity (Starlink/VSAT) for call centre location, typically co-located with shelters or NDMA Emergency Operations Centres. (2) Call centre uses text messaging (SMS) as primary channel to avoid network congestion, supplemented by voice for urgent queries. (3) Call centre staff are briefed on current SITREP content, health services, shelter locations, food distribution points, and family reunification procedures. (4) MNO coordination: request toll-free emergency number activation or SMS shortcode per NETP Action Item 3 provisions. (5) Call centre activity logs are fed into SOP 6.3 data collection for SITREP production. (6) International partner support (WFP ETC, IFRC) may be requested for call centre capacity augmentation via SHOC Telecom SOP v3.0 surge mechanisms. Annex 6.3-J provides the Call Centre Setup Checklist and staffing template.

7. SOP 6.4: Business Continuity and Disaster Recovery

Purpose: Operationalise ICT Policy v3.0 Section 12 by providing step-by-step procedures for three-tier failover activation, disaster recovery testing, SPOF mitigation verification, power continuity, and BCM Plan (SOP 9.1–9.7) interface. This SOP ensures SHOC survives its own ICT disruptions — not just responds to external disasters.

Linked Annexes: 6.4-A (Three-Tier Failover Checklist), 6.4-B (Botswana OSOCC Readiness Verification), 6.4-C (Virtual OSOCC Provisioning Checklist), 6.4-D (DR Test Plan Template), 6.4-E (Cloud Restore Test Record), 6.4-F (SPOF Testing Schedule), 6.4-G (Power Continuity Test Log), 6.4-H (Post-Recovery Validation Checklist), 6.4-I (BCM-ICT Joint Activation Checklist)



7.1 Three-Tier Failover Procedures

#	Action	Responsible	Output	Threshold	Annex Ref
F1	Automated monitoring detects threshold breach (connectivity loss >5 min, EW feed latency >15 min, power failure). Classify severity: Minor (single component), Major (multiple components), Critical (total site loss).	ICT Officer	Incident Classification	Automatic	6.4-A
F2	Tier 1 (Nacala on-site): Activate redundant path — secondary ISP (<30s), Starlink satellite (<5 min), or generator/UPS for power events. Verify EW feeds, DRMIMS, AMHEWAS restored.	ICT Officer	Tier 1 Active	<30 sec–5 min	6.4-A
F3	Tier 2 (Botswana OSOCC): If Tier 1 fails after 30 min or site evacuated: activate satellite uplink, redirect EW feeds via AMHEWAS redundant node, sync DRMIMS to Botswana buffer. Notify ED, SADC Secretariat, NDMAs.	ICT Officer + Head Ops	Tier 2 Active	<2 hours	6.4-B
F4	Tier 3 (Virtual OSOCC): If Botswana OSOCC also unavailable: activate Azure cloud environment, enable VPN remote access for all critical staff, switch to cloud-based EW/IM/Ops. Notify Governing Board.	ED + ICT Officer	Tier 3 Active	<4 hours	6.4-C
F5	Post-recovery: validate data integrity per classification level (Public: checksum; Confidential+: hash + audit trail; Secret+: ED verification). Restore primary site. File AAR within 7 days.	ICT Officer	Recovery Complete	<72 hours	6.4-H

7.2 DR Testing Schedule

#	Action	Responsible	Output	Threshold	Annex Ref
T1	Monthly: backup integrity verification — automated checksum of all backup sets; restore test of one random system per month; RPO compliance verification for all categories.	Asst ICT	Backup Test Report	Monthly	6.4-D
T2	Quarterly: cloud restore drill — restore DRMIMS + EW feeds to Azure secondary region; verify data consistency within RPO; measure actual RTO against target; document variance.	ICT Officer	Cloud Restore Record	Quarterly	6.4-E
T3	Quarterly: satellite failover drill — simulate primary ISP loss; verify auto-	ICT Officer	Failover Drill Report	Quarterly	6.4-A

	failover to Starlink within 30 sec; measure EW feed continuity; test BGAN backup.				
T4	Quarterly: Botswana OSOCC readiness — verify satellite uplink, DRMIMS sync lag <15 min, data buffer integrity, physical equipment status, access credentials current.	ICT Officer	OSOCC Readiness Report	Quarterly	6.4-B
T5	Semi-annual: SPOF testing — systematically disable each SPOF mitigation and verify failover per ICT Policy §12.3 table. Document results and remediation actions.	ICT Officer	SPOF Test Report	Semi-annual	6.4-F
T6	Annual: full BCM exercise with BCM Plan (SOP 9.1–9.7) — simulate Critical site-loss scenario; activate Tier 2/3; operate for minimum 4 hours; conduct joint AAR with BCM team.	ED + Head Ops + ICT	BCM Exercise Report	Annual	6.4-I

7.3 Power Continuity

Monthly: UPS battery load test (30 min sustained); verify automatic transfer switch (<10 sec). Quarterly: generator full-load test (2 hours); verify fuel reserve ≥72 hours at full load; test solar panel output against rated capacity. Annual: UPS battery replacement assessment; generator service per manufacturer schedule. All tests documented in Power Continuity Test Log (Annex 6.4-G). ICT Officer maintains fuel reserve monitoring with automated alerts at 50% and 25% levels.

7.4 Recovery Procedures by Classification

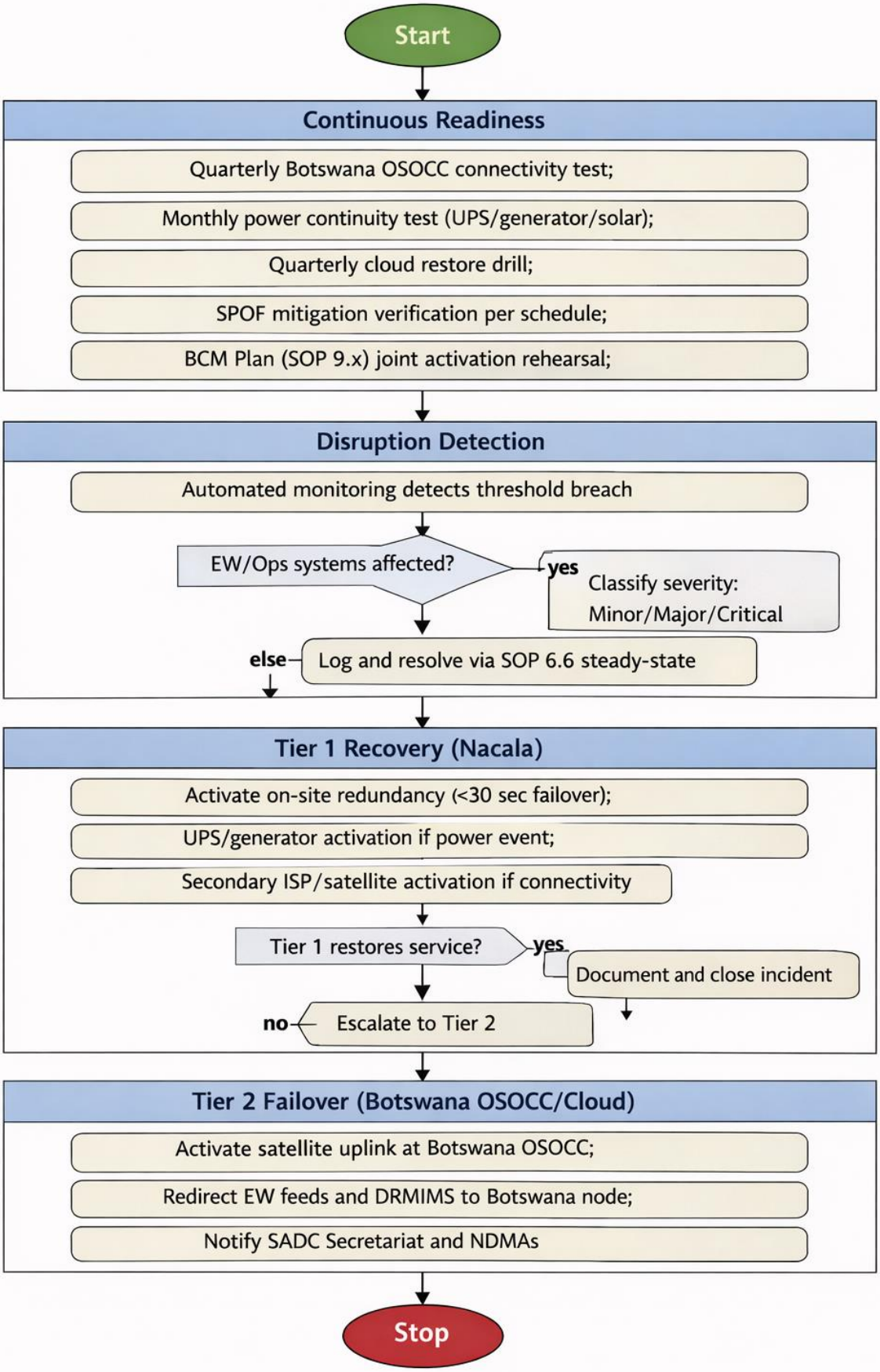
Public/Restricted data: restore from latest backup; verify checksums; resume operations. Confidential data: restore from encrypted backup; verify hash integrity and audit trail continuity; ICT Officer signs off. Secret data: restore from encrypted backup with restricted access restore environment; verify hash + audit trail + access log continuity; Head of Operations signs off. Top Secret data: restore under ED authorisation only; restricted access restore environment with enhanced monitoring; ED physical sign-off. All classification-specific recovery follows Annex 6.4-H (Post-Recovery Validation Checklist).

8. SOP 6.5: Cybersecurity Operations

Purpose: Operationalise ICT Policy v3.0 Section 10 by providing continuous cybersecurity procedures for incident response, vulnerability management, penetration testing, security monitoring, endpoint protection, patch management, security training, and compliance auditing. Ensures SHOC maintains a proactive security posture 24/7, not only during emergency activations.

Linked Annexes: 6.5-A (Incident Response Plan), 6.5-B (Incident Severity Classification Matrix), 6.5-C (Vulnerability Scanning Schedule & Remediation Tracker), 6.5-D (Penetration Test Scoping Template), 6.5-E (SIEM Alert Triage Procedures), 6.5-F (Patch Management Lifecycle), 6.5-G (Endpoint Security Enrolment Checklist), 6.5-H (Security Awareness Training Programme), 6.5-I (Security Audit Preparation Checklist), 6.5-J (DLP Policy and Controls)

SOP 6.4: BCM & Disaster Recovery Lifecycle



8.1 Incident Response

#	Action	Responsible	Output	Threshold	Annex Ref
IR1	Detection: SIEM/SOC alert or user report triggers incident. Classify per severity matrix: P1 (Critical: data breach, ransomware, EW system compromise), P2 (High: targeted attack, privilege escalation), P3 (Medium: malware, phishing success), P4 (Low: policy violation, suspicious activity).	ICT Officer	Incident Ticket	<15 min	6.5-A, 6.5-B
IR2	Containment: P1/P2: isolate affected systems immediately; disable compromised accounts; preserve forensic evidence (disk image, logs). P3/P4: quarantine endpoint; block indicators of compromise. Notify ED for P1; Head Ops for P2.	ICT Officer	Containment Report	P1: <1hr; P2: <4hr	6.5-A
IR3	Eradication: remove malware/backdoors; patch vulnerability exploited; harden configuration; rotate credentials for affected accounts. External forensics for P1 incidents.	ICT Officer + SECaaS	Eradication Report	<24 hours (P1/P2)	6.5-A
IR4	Recovery: restore from verified clean backup; validate data integrity per SOP 6.4 classification procedures; resume services with enhanced monitoring for 72 hours. Confirm no re-infection.	ICT Officer	Recovery Confirmation	Per RTO targets	6.5-A, 6.4-H
IR5	Lessons learned: AAR within 7 days; update SIEM rules; update training materials; report to SADC Secretariat ICT Division if cross-border implications. Semi-annual IR drill to test full lifecycle.	ICT Officer	IR AAR Report	7 days post	6.5-A

8.2 Vulnerability and Patch Management

Vulnerability Scanning (Weekly): automated scanning of all externally-facing and internal systems using approved tools (Nessus/Qualys/equivalent). Results triaged by severity: Critical vulnerabilities remediated within 24 hours; High within 72 hours; Medium within 14 days; Low within 30 days. Monthly vulnerability posture report to Head of Operations. Remediation tracked in Annex 6.5-C.

Penetration Testing (Annual): external pen test scoped per Annex 6.5-D covering: external perimeter, web applications, satellite communications, VPN endpoints, cloud infrastructure. Results remediated per severity timeline above. Re-test of Critical/High findings within 30 days.

Patch Management (Monthly cycle + Emergency): standard patches tested in staging environment for 48 hours then deployed. Emergency patches (zero-day, actively exploited) deployed within 24 hours with rollback plan. Patch lifecycle documented in Annex 6.5-F. Rollback procedures: snapshot before deployment; automated rollback trigger if critical service degraded within 2 hours.

8.3 Security Monitoring and Endpoint Protection

SIEM Alert Triage: 24/7 SOC (SECaaS provider) monitors SIEM. Alert triage procedures per Annex 6.5-E: Critical alerts escalated to ICT Officer within 15 min; High within 1 hour; Medium batched daily; Low batched weekly. False positive rate tracked and SIEM rules tuned quarterly targeting <5% false positive rate.

Endpoint Security: all SHOC devices enrolled in MDM per Annex 6.5-G. Compliance checks: device encryption enabled, EDR agent active, OS patches current, no unapproved software. Non-compliant devices quarantined from network within 24 hours. Lost/stolen device procedure: remote wipe within 1 hour of report; credential rotation for user; incident logged.

Data Loss Prevention: classified data (Confidential+) protected by DLP controls per Annex 6.5-J. USB storage disabled on all endpoints except authorised devices. Email DLP rules flag outbound attachments containing classified keywords. Cloud DLP monitors data exfiltration patterns. Monthly DLP incident review with Head of Operations.

8.4 Security Training and Audit

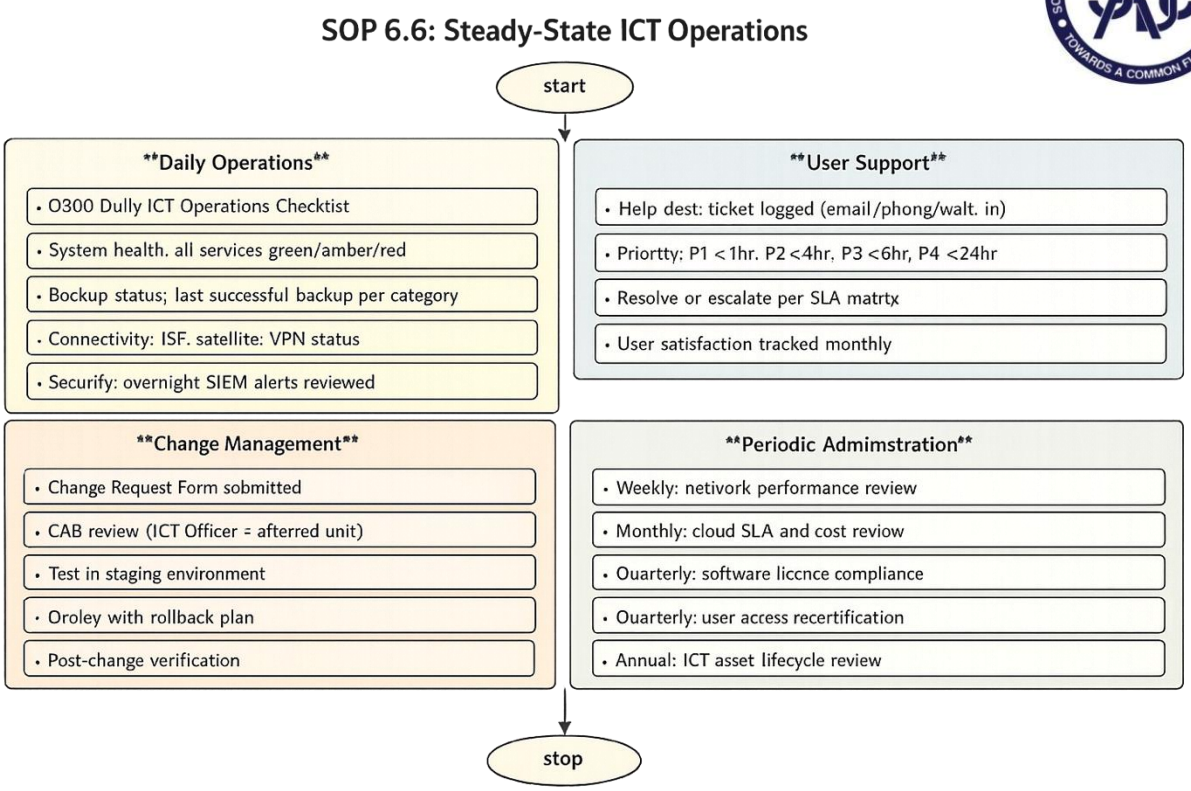
Quarterly Security Awareness Training per Annex 6.5-H: phishing simulation (target <10% click rate), password hygiene, social engineering awareness, data classification handling, incident reporting procedures. New staff: mandatory completion within first week. Annual training effectiveness assessment with programme updates.

Annual Security Audit per Annex 6.5-I: external auditor conducts ISO 27001 gap assessment covering ISMS documentation, access controls, encryption, monitoring, incident response, BCM, and supplier management. Findings tracked with remediation timeline. Management review of audit results within 30 days. Compliance evidence maintained in ISMS document register.

9. SOP 6.6: Steady-State ICT Operations

Purpose: Provide operational procedures for day-to-day ICT administration, user support, change management, network monitoring, asset lifecycle management, and cloud service governance. Ensures SHOC's ICT environment is professionally managed 24/7, not only during emergency activations. Operationalises ICT Policy Sections 10–16.

Linked Annexes: 6.6-A (Daily ICT Operations Checklist), 6.6-B (Help Desk Ticket Form & SLA Matrix), 6.6-C (Change Request Form & CAB Procedures), 6.6-D (Network Monitoring Dashboard Config), 6.6-E (Cloud SLA Monitoring Template), 6.6-F (User Account Lifecycle Checklist), 6.6-G (Software Licence Register), 6.6-H (ICT Asset Lifecycle Register), 6.6-I (ICT Succession & Cross-Training Plan)



9.1 Daily Operations and Help Desk

Daily ICT Operations Checklist (Annex 6.6-A) executed by Asst ICT at 0800 daily: verify all systems operational (EW feeds, DRMIMS, ERP, HRIS, AMHEWAS, VOSOCC, cloud services); confirm backup status for all categories; check ISP/satellite/VPN connectivity; review overnight SIEM alerts; confirm UPS/generator status; log any anomalies. Checklist signed by ICT Officer. Escalate any RED items to ICT Officer immediately.

Help Desk (Annex 6.6-B): all ICT requests logged via email/phone/walk-in. Ticket assigned priority: P1 (critical system down, <1hr response), P2 (degraded service, <4hr), P3 (non-urgent, <8hr), P4 (enhancement, <24hr). SLA compliance tracked monthly. Unresolved P1/P2 tickets escalated to ICT Officer after initial response window. Monthly help desk report to Head of Operations.

9.2 Change Management

All ICT changes (infrastructure, software, configuration, access) require a Change Request Form (Annex 6.6-C). Standard changes (pre-approved, low risk): execute per documented procedure. Normal changes: submit form → CAB review (ICT Officer + affected functional unit head) → test in staging → deploy with rollback plan → post-change verification within 24hr. Emergency changes (critical security/operations): ICT Officer authorises immediately; document retrospectively within 48hr; CAB review at next meeting. All changes logged in change register with rollback procedures documented.

9.3 Network and Cloud Monitoring

Network Monitoring (Annex 6.6-D): tools (PRTG/Zabbix/equivalent) monitor all links (ISP, satellite, VPN, LAN). Thresholds: latency >100ms amber, >500ms red; packet loss >1% amber, >5% red; bandwidth utilisation >80% amber, >95% red. Automated alerts to ICT Officer for red events. Weekly network performance report. Monthly bandwidth utilisation trending for capacity planning.

Cloud SLA Monitoring (Annex 6.6-E): monthly review of Azure SLA compliance (≥99.9% uptime), cost vs. budget, storage utilisation, and egress charges. Capacity planning review: project 6-month growth; adjust resource allocation. Annual cloud provider assessment per ICT Policy §14.2 criteria.

9.4 User Account Lifecycle (Gap D7)

Onboarding (Annex 6.6-F): HR notifies ICT of new hire ≥ 5 days before start. Day 1 setup: create AD/Azure accounts; assign RBAC per role; provision email, DRMIMS, ERP access; configure device (MDM enrolled, encrypted); issue credentials securely; deliver ICT orientation (security awareness, acceptable use, help desk contacts). Role changes: access modification within 24hr of HR notification; remove old permissions before adding new. Offboarding: same-day deactivation of all accounts upon departure notification; device collection and wipe; transfer of data ownership; final access audit. Quarterly: ICT Officer conducts access recertification with each functional unit head — verify all active accounts are current staff with appropriate access levels.

9.5 Asset Lifecycle and Licensing (

ICT Asset Lifecycle (Annex 6.6-H): asset register maintained in ERP with acquisition date, warranty, assigned user, condition, and refresh schedule. Hardware refresh: laptops/tablets 4-year cycle; servers 5-year; network equipment 7-year; satellite equipment per manufacturer recommendation. Warranty tracking: alert at 60 days before expiry. Secure disposal: data-classified assets wiped per ICT Policy §13.1 then physically destroyed (Certificate of Destruction for Confidential+).

Software Licensing (Annex 6.6-G): register maintained for all licensed software (OS, productivity, security, specialist). Quarterly compliance check: verify licences cover active installations. Renewal tracking: alert at 90 days before expiry. Annual licence optimisation: review usage data and right-size subscriptions.

9.6 Succession Planning and ICT Policy Review

ICT Succession Plan (Annex 6.6-I): ICT Officer → Asst ICT documented succession. Quarterly skill rotation: Asst ICT performs ICT Officer duties for 1 week per quarter under supervision. Knowledge transfer register maintained for all critical ICT functions (satellite activation, cloud administration, SIEM management, DRMIMS admin, BCM failover). Emergency cover: if ICT Officer unavailable, Asst ICT assumes all duties; if both unavailable, Head of Operations activates WFP ETC or IFRC IT surge.

ICT Policy Review (per Policy §18): every 24 months — ICT Officer leads review with stakeholder consultation (all functional unit heads); gap analysis against current technology environment and compliance requirements; draft revisions circulated for 30-day comment; ED approval. SOP review follows same cycle aligned to Policy review.

— END OF SOPs: ICT, DATA MANAGEMENT AND REPORTING —

Next Review Date: 31 December 2027

Risk Matrix

This single risk matrix consolidates risks from all three SOPs. Scored on a 5×5 likelihood impact model with colour-coded severity levels and predefined action thresholds.

#	Risk	L	I	Rating	Mitigation
1	Data Breach / Information Security	2	5	10 Amber	MFA; encryption; SOC monitoring; staff training; ISO 27001 controls
2	Cyber Attack	3	5	15 Red	SECaaS; EDR; firewall; SOC 24/7; TESS+ baseline; incident response
3	Local MNO / Connectivity Failure	3	4	12 Amber	Multi-carrier redundancy; Starlink/VSAT backup; dual submarine cables
4	Power Outage (Nacala)	4	4	16 Red	UPS; dual generator; solar backup; cloud failover to Azure
5	System / Dashboard Failure	2	5	10 Amber	Cloud failover; geo-redundant; 2-hour RTO; DRaaS
6	Data Corruption / Quality Issues	2	4	8 Yellow	Validation scripts; version control; hourly backups; QA protocols
7	Satellite Capacity Saturation	3	3	9 Yellow	Multi-provider; Crisis Connectivity Charter; bandwidth management
8	Partner Data Feed Loss	3	4	12 Amber	Redundant feeds; backup sources; manual fallback; AMHEWAS nodes
9	SITREP Production Delay	3	4	12 Amber	Template automation; shift coverage; backup IM staff; surge roster
10	Regional VPN Failure	2	4	8 Yellow	Redundant tunnels; alternative routing; web portal backup
11	ETK Equipment Failure	2	4	8 Yellow	Redundant equipment; pre-positioned spares; SLA support
12	Backup / Recovery Failure	2	5	10 Amber	Multi-cloud; automated verification; alerts; monthly DR tests
13	Unpatched Vulnerability Exploited	3	5	15 Red	Weekly scanning; Critical <24hr patch; pen testing; SIEM monitoring
14	Insider Threat / Data Exfiltration	2	5	10 Amber	DLP controls; USB disable; email DLP; RBAC; quarterly access review
15	Three-Tier Failover Failure	2	5	10 Amber	Quarterly failover drills; OSOCC readiness tests; cloud restore drills
16	Phishing / Social Engineering	3	4	12 Amber	Quarterly training; phishing simulations; email filtering; MFA enforcement
17	ICT Officer Single Point of Failure	3	4	12 Amber	Succession plan; quarterly skill rotation; cross-training programme
18	Change Management Failure	2	4	8 Yellow	CAB process; staging testing; rollback procedures; post-change verification

BGYOR Action Thresholds: Blue (1–4): Standard monitoring, quarterly review. Green (5–8): Unit-level response, weekly reporting. Yellow (9–12): HoP notified, daily monitoring, pre-positioned resources. Amber (13–16): ED notified, BCM protocols, GB informed 48hrs. Red (17–25): Immediate ED+GB intervention, full BCM activation, external audit, expenditure frozen.

Cross-Cutting SOP Integration

This single table replaces all per-section cross-cutting references. It defines the mandatory integration points between Volume 6 and other SHOC SOP Volumes.

SOP Volume	Reference	Integration with Vol 6	Key Interface Actions
Vol 1: Finance	SOP 1.1–1.4	Financial data management; ERP system infrastructure; EPF tracking data	IFMS data feeds → DRMIMS; financial dashboards; audit trail data; donor report inputs
Vol 2: Early Warning	SOP 2.1–2.3	EW data feeds; AMHEWAS/GDACS integration; CAP message processing and validation	Real-time hazard data ingestion; EW dashboard; CAP validation per Step 2.3A; threshold monitoring; SITREP Section 2
Vol 3: Emergency Comms / Telecom SOP v3	SOP 3.1–3.2 + Telecom SOP	ETK deployment; PPDR frequency management; Tampere Convention facilitation; customs clearance for equipment; Cell Broadcast/CAP Public Warning; quarterly telecom readiness drills	ETK deployed with ERT per PPDR bands; SATCOM activation; Frequency Pre-Clearance Cache; customs document packs; CAP approval chains; CB/LB-SMS fallback; drill schedule integration with Vol 6 ICT testing
Vol 4: ERT	SOP 4.1–4.4	ERT technology packages; field data collection; OSOCC integration	Field tablet data → DRMIMS; ERT status tracking; coordination logs; SITREP Section 3
Vol 5: HR	SOP 5.1–5.2	HRIS platform; personnel data management; surge roster systems	HR data classification (Confidential); welfare tracking; SITREP Section 6
Vol 7: Security	SOP 7.1–7.3	Access control systems; CCTV; SIEM; cybersecurity monitoring	Access logs to DRMIMS; security incident data; SITREP Section 8
Vol 8: SCM	SOP 8.1–8.5	Procurement systems; asset tracking; supplier database	Procurement data feeds; asset registry; logistics dashboard; SITREP Section 4
Vol 9: BCM	SOP 9.1–9.7	Backup/DR infrastructure; failover procedures; continuity architecture	RPO/RTO targets drive backup schedules; failover triggers; BCM register data
SADC NETP / PPDR	NETP (2023); PPDR (2021)	NETP domestication tracking via MS Readiness Matrix (Telecom SOP); PPDR frequency harmonisation; Tampere Convention; EW4All; SADC ET Directory	NETP focal point registers; PPDR band alignment for all ETK equipment; customs facilitation per Tampere; accessibility per AI 10; call centre procedures per AI 14
Vol 9: BCM Plan	SOP 9.1–9.7	BCM failover architecture; continuity triggers; joint activation procedures	SOP 6.4 operationalises three-tier failover; BCM-ICT joint activation checklist; annual full BCM exercise; SPOF testing aligned to BCM risk register

Key Performance Indicators

Indicator	Target	Frequency	Owner	SOP Source
Data Quality Score (≥ validation pass rate)	≥95%	Weekly	IM Officer	6.1
Backup Success Rate	100%	Daily	ICT Officer	6.1
RPO Compliance (EW data)	≤15 min	Continuous	Asst ICT	6.1/6.2
Classification Compliance	≥98%	Monthly	ICT Officer	6.1
Network Availability	≥99.5%	Continuous	ICT Officer	6.2
SATCOM Readiness	100%	Monthly	ICT Officer	6.2
Cybersecurity Incident Response	<4 hours	Per incident	ICT Officer	6.2
SITREP Timeliness	≥95%	Per event	IM Officer	6.3
Dashboard Uptime	≥99.9%	Continuous	ICT Officer	6.3
Partner Data Latency	<15 min	Per event	IM Officer	6.3
Report Accuracy (corrections needed)	<2%	Monthly	Head of Ops	6.3
Cross-Functional Integration	100%	Quarterly	ICT Officer	All
CAP Message Validation Pass Rate	≥98%	Per event	IM Officer	6.1
Accessibility Compliance (SITREP/Flash)	100%	Per publication	IM Officer	6.3
PPDR Band Alignment (ETK equipment)	100%	Annual	ICT Officer	6.2
Failover Drill Success Rate	100%	Quarterly	ICT Officer	6.4
Cloud Restore Test RTO Compliance	100%	Quarterly	ICT Officer	6.4
Botswana OSOCC Readiness Verified	100%	Quarterly	ICT Officer	6.4
Power Continuity Test Pass Rate	100%	Monthly	Asst ICT	6.4
SPOF Mitigations Verified	100%	Semi-annual	ICT Officer	6.4
Vulnerability Scanning (weekly)	100%	Weekly	ICT Officer	6.5
Critical Vuln Remediation <24hr	≥95%	Per event	ICT Officer	6.5
Security Awareness Training	100% staff	Quarterly	ICT Officer	6.5
Phishing Simulation Click Rate	<10%	Quarterly	ICT Officer	6.5
IR Drill Completion	100%	Semi-annual	ICT Officer	6.5
Patch Compliance (30-day)	≥95%	Monthly	Asst ICT	6.5
Daily Ops Checklist Completion	100%	Daily	Asst ICT	6.6
Help Desk SLA Compliance	≥90%	Monthly	Asst ICT	6.6

User Account Lifecycle <24hr	100%	Per event	ICT Officer	6.6
Software Licence Compliance	100%	Quarterly	ICT Officer	6.6

Terms and Abbreviations

Abbreviation	Full Term
ACMAD	African Centre of Meteorological Application for Development
AMHEWAS	Africa Multi-Hazard Early Warning and Action System
AU	African Union
BCM	Business Continuity Management
BGYOR	Blue, Green, Yellow, Orange (Amber), Red – Risk rating framework
BGAN	Broadband Global Area Network
CAB	Change Advisory Board – reviews and approves IT changes per SOP 6.6
CAP	Common Alerting Protocol – standardised alerting format per OASIS/ITU standards
CB/LB-SMS	Cell Broadcast / Location-Based SMS – geo-located emergency alert technology per NETP Action Item 8
CRASA	Communications Regulators' Association of Southern Africa
DLP	Data Loss Prevention – controls preventing exfiltration of classified data per SOP 6.5
DRMIMS	Disaster Risk Management Information Management System
DRaaS	Disaster Recovery as a Service
ED	Executive Director (SHOC)
EDR	Endpoint Detection and Response – advanced endpoint security per SOP 6.5
ERT	Emergency Response Team
ETC	Emergency Telecommunications Cluster (WFP)
ETK	Emergency Telecommunications Kit
EW	Early Warning
EW4All	Early Warning for All – UN initiative for universal multi-hazard EWS; four pillars including Warning Dissemination (Pillar 3)
GDACS	Global Disaster Alert and Coordination System
GIS	Geographic Information System
GIS/DCM	GIS and Disaster Connectivity Mapping – geospatial layers for telecom coverage, vulnerability, and logistics nodes
GTS	Global Telecommunication System (WMO)

HDX	Humanitarian Data Exchange
HoP	Head of Programmes
HRIS	Human Resource Information System
IFRC	International Federation of Red Cross and Red Crescent Societies
IM	Information Management
IR	Incident Response – cybersecurity incident lifecycle per SOP 6.5
ISMS	Information Security Management System – per ISO 27001:2022
INAM	Instituto Nacional de Meteorologia (Mozambique)
INGD	Instituto Nacional de Gestão e Redução do Risco de Desastres (Mozambique)
LEO	Low Earth Orbit
MDM	Mobile Device Management – endpoint device governance per SOP 6.5
MEL	Monitoring, Evaluation and Learning
MNO	Mobile Network Operator
NDMA	National Disaster Management Authority
NETP	National Emergency Telecommunications Plan (SADC Model, October 2023)
NETWG	National Emergency Telecommunications Working Group – multi-stakeholder body per NETP
NRA	National Regulatory Authority (telecommunications) – per NETP Action Item 1
OSOCC	On-Site Operations Coordination Centre
PAM	Privileged Access Management
PPDR	Public Protection and Disaster Relief – SADC harmonised RF framework (2021)
PWD	Persons With Disabilities
RBAC	Role-Based Access Control
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SADC	Southern African Development Community
SATCOM	Satellite Communications
SAWS	South African Weather Service
SECaaS	Security as a Service
SHOC	SADC Humanitarian and Emergency Operations Centre

SIEM	Security Information and Event Management
SITREP	Situation Report
SOC	Security Operations Centre
SPOF	Single Point of Failure – identified and mitigated per SOP 6.4 and ICT Policy §12.3
Tampere Convention	Convention on the Provision of Telecommunication Resources for Disaster Mitigation and Relief Operations (1998)
TESS+	UN Telecommunications Security Standards
TETRA/DMR/P25	Terrestrial Trunked Radio / Digital Mobile Radio / Project 25 – preferred PPDR equipment standards
UNDAC	United Nations Disaster Assessment and Coordination
VOSOCC	Virtual On-Site Operations Coordination Centre
VSAT	Very Small Aperture Terminal
WFP	World Food Programme
WMO	World Meteorological Organization

References

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WFP FITTEST Deployment Protocols.

UN TESS+ Telecommunications Security Standards.

Consolidated Annexes Register

Annex	Title	SOP	Phase(s)
6.1-A	Data Receipt Register Template	6.1	All
6.1-B	Metadata Schema and Tagging Guide (DRMIMS)	6.1	All
6.1-C	Data Classification Matrix (by Functional Area)	6.1	All
6.1-D	Data Request Register Template	6.1	All
6.1-E	Backup Verification Checklist	6.1	All
6.1-F	Data Quality Assessment Form	6.1	All
6.1-G	Regional/Global Data Exchange Specifications	6.1	All
6.1-H	Cross-Functional Data Flow Diagram (SOPs 1–9)	6.1	All
6.1-I	Data Stewardship Responsibilities Matrix	6.1	All
6.1-J	After-Action Review Template	6.1	Demobilisation
6.2-A	Mozambique MNO Contact Directory and SLA Summary	6.2	All
6.2-B	Satellite Equipment Inventory and Deployment Checklist	6.2	All
6.2-C	Regional VPN Configuration Standards	6.2	All
6.2-D	Cloud Service Evaluation Matrix	6.2	Preparedness
6.2-E	Security Baseline Configuration (TESS+ Aligned)	6.2	All
6.2-F	AU/UN Platform Access Credentials Register	6.2	All
6.2-G	MET Service Data Feed Specifications	6.2	All
6.2-H	ETC Coordination Procedures and Contacts	6.2	Activation/Ops
6.2-I	ETK Deployment Checklist	6.2	Activation/Ops
6.2-J	Incident Response Procedures	6.2	All
6.2-K	Network Topology Diagrams	6.2	All
6.2-L	Cross-Functional ICT Requirements Matrix	6.2	Preparedness
6.2-M	SADC PPDR Harmonised Frequency Band Reference and Equipment Compatibility Matrix	6.2	All Phases
6.3-A	SITREP Template	6.3	Activation/Ops
6.3-B	Flash Update Template	6.3	Alert/Activation

6.3-C	Rapid Needs Assessment Form	6.3	Alert/Activation
6.3-D	Dashboard Indicator Definitions	6.3	All
6.3-E	Data Quality Checklist	6.3	All
6.3-F	Cross-SOP Data Flow Diagram	6.3	All
6.3-G	AAR Template	6.3	Demobilisation
6.3-H	Donor Reporting Template	6.3	Operations/Demob
6.3-I	Accessibility Checklist (PWD, multilingual, accessible formats)	6.3	All Phases
6.3-J	Call Centre Setup Checklist and Staffing Template	6.3	Operations
6.4-A	Three-Tier Failover Activation Checklist	6.4	All Phases
6.4-B	Botswana OSOCC Readiness Verification Checklist	6.4	Quarterly
6.4-C	Virtual OSOCC Provisioning Checklist	6.4	All Phases
6.4-D	DR Test Plan Template	6.4	Monthly/Quarterly
6.4-E	Cloud Restore Test Record	6.4	Quarterly
6.4-F	SPOF Testing Schedule and Results Log	6.4	Semi-annual
6.4-G	Power Continuity Test Log	6.4	Monthly
6.4-H	Post-Recovery Validation Checklist	6.4	Per event
6.4-I	BCM-ICT Joint Activation Checklist	6.4	Annual exercise
6.5-A	Incident Response Plan	6.5	All Phases
6.5-B	Incident Severity Classification Matrix	6.5	Per event
6.5-C	Vulnerability Scanning Schedule and Remediation Tracker	6.5	Weekly
6.5-D	Penetration Test Scoping Template	6.5	Annual
6.5-E	SIEM Alert Triage Procedures	6.5	Continuous
6.5-F	Patch Management Lifecycle	6.5	Monthly
6.5-G	Endpoint Security Enrolment Checklist	6.5	Per device
6.5-H	Security Awareness Training Programme	6.5	Quarterly
6.5-I	Security Audit Preparation Checklist	6.5	Annual
6.5-J	DLP Policy and Controls	6.5	Continuous
6.6-A	Daily ICT Operations Checklist	6.6	Daily
6.6-B	Help Desk Ticket Form and SLA Matrix	6.6	Continuous
6.6-C	Change Request Form and CAB Procedures	6.6	Per change
6.6-D	Network Monitoring Dashboard Configuration	6.6	Continuous

6.6-E	Cloud SLA Monitoring Template	6.6	Monthly
6.6-F	User Account Lifecycle Checklist	6.6	Per event
6.6-G	Software Licence Register	6.6	Quarterly
6.6-H	ICT Asset Lifecycle Register	6.6	Annual
6.6-I	ICT Succession and Cross-Training Plan	6.6	Quarterly

Document Version Control

Version	Author	Designation	Date	Status
1.0–3.0	I Janse van Rensburg	SOP Author	Dec 2025	SOPs 6.1–6.3 separate drafts
3.0+	IFRC Review	Reviewer	Feb 2026	IFRC feedback on SOP 6.2
4.0	ARC / Claude	Consolidation	Mar 2026	UNDAC Cycle alignment; 3-SOP merger
5.0	ARC / Claude	Benchmark Repair	Mar 2026	SADC NETP/PPDR gap remediation
6.0	ARC / Claude	Policy Compliance	Mar 2026	BCM/DR/Cyber/Steady-State SOPs added